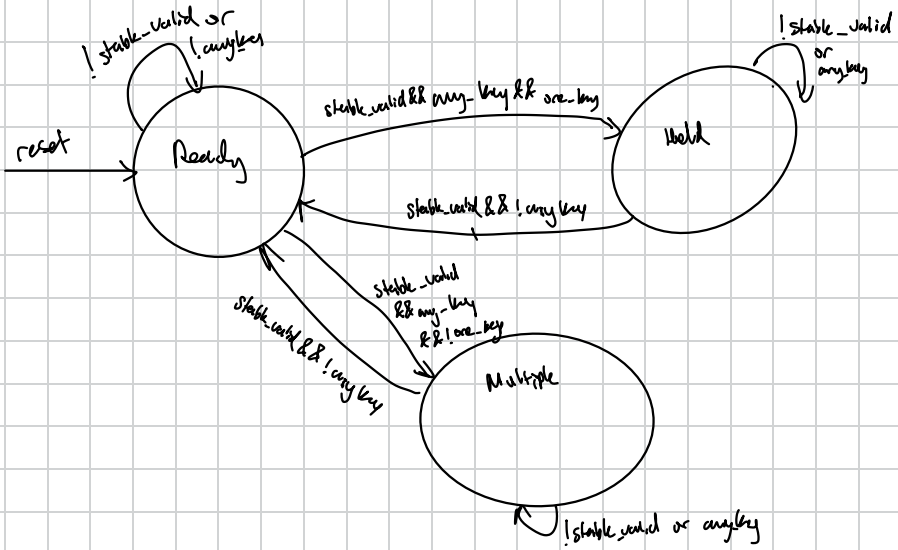
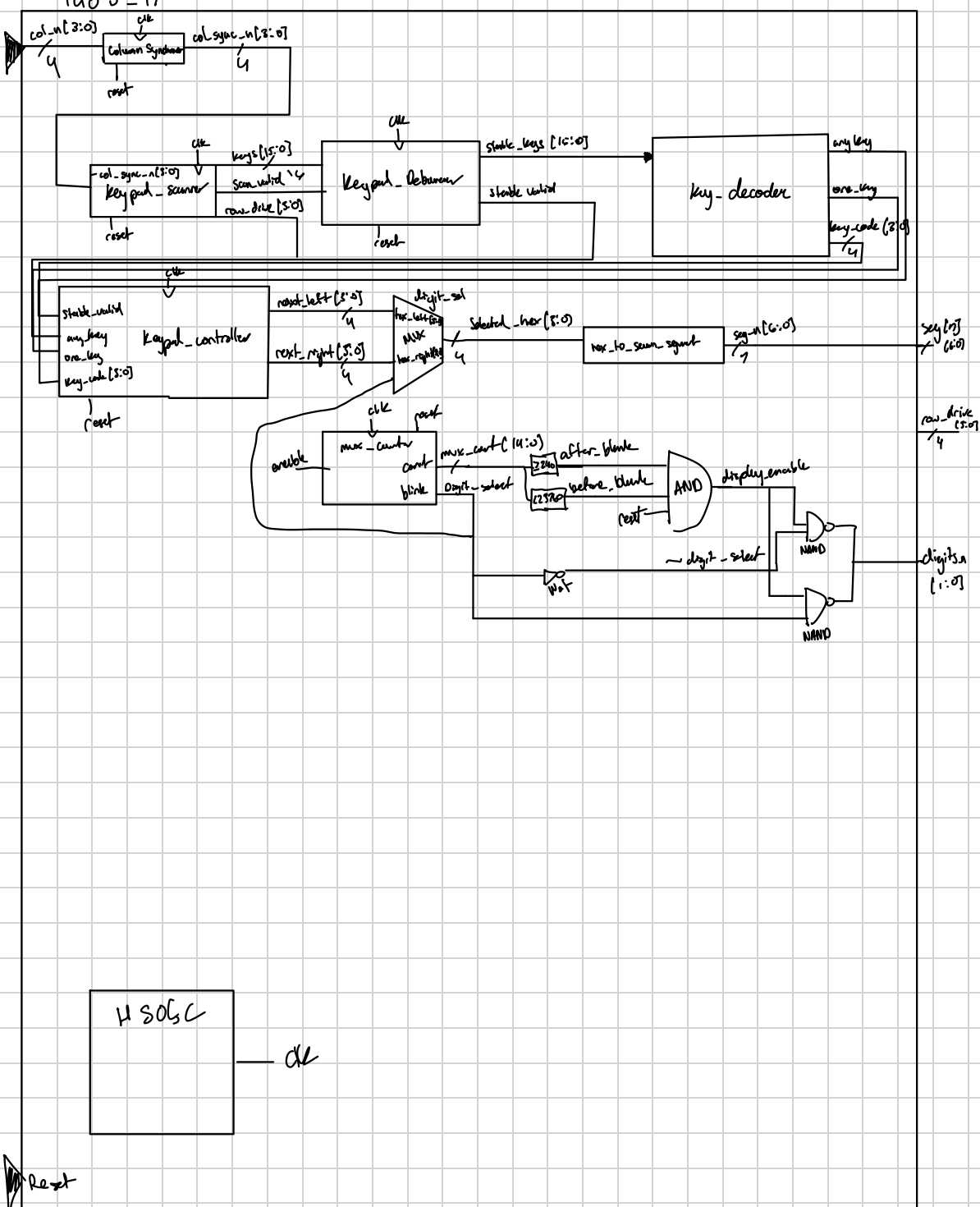
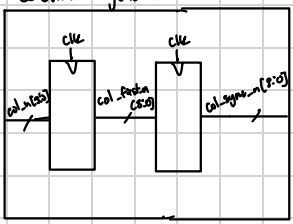


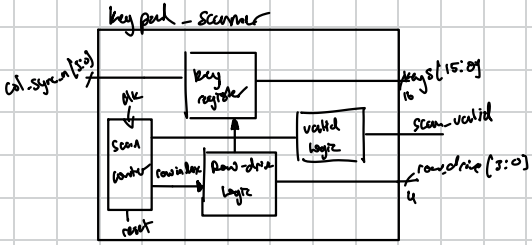

!stable_valid or !any_key
reset
Ready
stable_valid && any_key && one_key
Valid
!stable_valid or any_key
stable_valid && !any_key
stable_valid && any_key && !one_key
stable_valid && !any_key
Multiple
!stable_valid or any_key


lab 3_tr
col_in[3:0]
clk
Column Synchronizer
col_sync_in[3:0]
reset
col_sync_in[3:0]
Keypad_Scanner
keys[15:0]
scan_valid
row_drive[3:0]
Keypad_Debouncer
stable_keys[15:0]
stable valid
key_decoder
any key
one_key
key_code[3:0]
Keypad_controller
stable_valid
any_key
one_key
key_code[3:0]
reset_left[3:0]
reset_right[3:0]
digit_sel
MUX
selected_hex[3:0]
hex_to_seven_segment
seg_in[6:0]
seg[6:0]
row_drive[3:0]
enable
mux_counter
count
blink
mux_count[14:0]
digit_select
2240
after_blank
12582
before_blank
AND
display_enable
reset
NAND
NOT
~digit_select
digits_n[1:0]
HSOSC
clk
reset

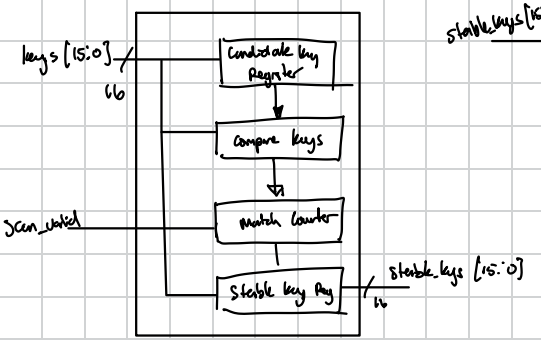
Column Synchronizer



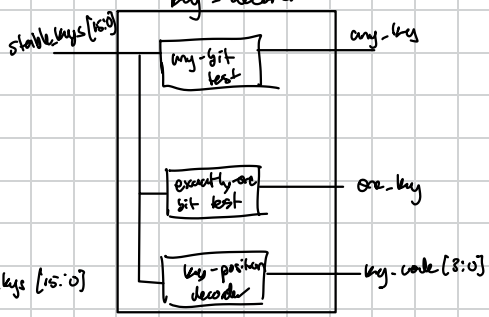
keypad_scanner



keypad_debouncer



key_decoder



keypad controller

